arXiv:2605.06415v1 [cs.LG] 7 May 2026

E = T·H/(O+B): A Dimensionless Control Parameter for Mixture-of-Experts Ecology

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May 8, 2026

Abstract

We introduce $E = T \cdot H/(O + B)$, a dimensionless control parameter that predicts whether Mixture-of-Experts (MoE) models will develop a healthy expert ecology or collapse into dead experts. E combines four hyperparameters—routing temperature T, routing entropy weight H, oracle weight O, and balance weight B—into a single quantity. Through 12 controlled experiments (8 vision, 4 language) totaling over 11,000 training epochs, we establish that $E \geq 0.5$ alone is sufficient to guarantee zero dead experts, removing the necessity for handcrafted load-balancing auxiliary losses. We validate this cross-modally on CIFAR-10, CIFAR-100, TinyImageNet-200, WikiText-2, and WikiText-103. Five additional findings emerge: (1) dead experts can *resuscitate*—contrary to traditional MoE theory where death is permanent, we observe recovery from 12 to 4 dead experts (E10→E80) on TinyImageNet-200; (2) ortho toxicity is dataset-dependent, not universal—CIFAR-10 and WikiText-2 show zero toxicity across all ortho values; (3) task complexity (number of classes) shifts the critical E threshold—TinyImageNet-200 at $E = 0.545$ produces DEAD=7 while CIFAR-10 at the same E maintains DEAD=0; (4) three-tier hierarchical MoE spontaneously collapses into a two-tier functional structure under high task complexity, with the middle tier permanently starved of routing flow; (5) ecological structure (tier usage, active expert count, hard-sample gradient) is temperature-invariant across a 50× range, while task accuracy is not—establishing ecological diagnostics as a structurally stable property decoupled from performance evaluation. Model overfitting is further shown to be decoupled from expert ecological health. We propose that E serves as a compact diagnostic for MoE training, analogous to the Reynolds number in fluid dynamics—a single dimensionless quantity that predicts regime transitions.

1 Introduction

Mixture-of-Experts (MoE) architectures [1, 2, 3] have become foundational to modern large-scale deep learning, powering systems from vision backbones [8] to GPT-4-scale language models. The core idea is elegant: route each input token to a subset of specialized expert sub-models, enabling conditional computation at scale without proportional increase in FLOPs.

Yet MoE training has a well-known pathology: *dead experts*. When the router consistently ignores certain experts, those experts receive no gradient signal, their representations degrade, and the feedback loop locks them out permanently. The standard response is load balancing—auxiliary losses that penalize uneven routing distributions [2, 1], minimum-usage constraints [4], or expert dropout [5]. More recently, DeepSeek proposed loss-free balancing via per-expert bias terms [6]. All of these treat the symptom (uneven routing) rather than the cause.

This paper proposes a different approach. Rather than fighting expert death with auxiliary constraints, we ask: *what causes a healthy expert ecology in the first place?*

We identify four hyperparameters that together govern the router's freedom to explore alternative expert assignments: routing temperature T, routing entropy weight H, oracle supervision weight O, and load balance weight B. We combine them into a single dimensionless quantity:

$$E = \frac{T \cdot H}{O + B} \tag{1}$$

E captures the net exploration budget available to the router. When E is high, the router has sufficient freedom to explore and discover which experts are competent at which inputs. When E is low, the router is constrained by strong oracle signals or aggressive load balancing, preventing the self-organization necessary for a healthy ecology.

Our contributions are:

1. **E as a unified control parameter.** We show that $E \geq 0.5$ alone is sufficient to guarantee zero dead experts across four datasets and two modalities. No auxiliary load-balancing loss is needed.
2. **Cross-modal validation.** We verify the E framework on both vision (CIFAR-10, CIFAR-100, TinyImageNet-200) and language (WikiText-2, WikiText-103) with a GPT-2 style MoE Transformer.
3. **Expert revival.** We document that dead experts can resuscitate—DEAD count drops from 12 to 4 (E10→E80) on TinyImageNet-200—contradicting the traditional MoE view that dead experts are permanently lost.
4. **Task complexity shifts the critical threshold.** 200-class TinyImageNet at $E = 0.545$ produces DEAD=7, while 10-class CIFAR-10 at the same E has DEAD=0. Task complexity is a missing variable in the E formula.
5. **Overfitting is decoupled from ecology.** WikiText-103 shows severe perplexity overfitting with zero expert collapse, demonstrating that ecological health and model generalization are independent axes.
6. **Ortho toxicity is dataset-dependent.** Three ortho sweeps across CIFAR-100, CIFAR-10, and WikiText-2 reveal that ortho toxicity is not universal—it manifests only on certain datasets at certain E values.
7. **Three-tier hierarchy collapses to two-tier.** Under high task complexity (TinyImageNet-200), the middle tier (T1) is permanently starved of routing flow, with usage dropping to 2.2%—the router spontaneously eliminates the redundant architectural tier.
8. **Ecological structure is temperature-invariant.** A 50× temperature scan (0.1–5.0) shows zero change in tier usage or active expert count, establishing that ecological diagnostics are structural properties independent of evaluation temperature. The T0/T2 hard-sample gradient persists at 1.2–1.6× across all temperatures.
9. **Routing lockdown as collapse mechanism.** The router enters a near-deterministic assignment state with 65.3% T0 self-loop rate, preventing cross-tier sample flow—this explains why raising E cannot fix task-complexity-induced collapse.

2 Related Work

Load balancing in MoE. The standard approach to dead experts is load balancing. [1] introduced auxiliary balance losses measuring the KL divergence between routing distribution and uniform distribution. [2] simplified to top-1 routing with capacity constraints. [4] proposed minimum-usage penalties and expert choice routing. [5] explored expert dropout and regularization schedules. [6] proposed per-expert bias terms that dynamically adjust to maintain balance without auxiliary loss gradients.

A common thread runs through all of these: they *enforce* uniform routing. Our findings suggest this may be counterproductive. Forcing uniform routing can homogenize experts—up to 99% output similarity has been observed in well-balanced MoE models. The alternative we present is to *allow* the ecosystem to self-organize, provided E is sufficient.

Orthogonality in MoE. [7] (NeurIPS 2025 Oral) propose an orthogonality loss on per-token expert outputs to advance expert specialization, reporting +23.79% relative improvement. Our work qualifies this finding: we show that orthogonality's effect on expert health is mediated by E. At low E, ortho amplifies winner-take-all dynamics and increases dead expert count. At healthy E, ortho is tolerated and may be modestly beneficial. We recommend that future work incorporating orthogonality losses report both E and per-expert health metrics.

Emergent specialization. [10] observed expert specialization along domain boundaries in language models. [11] found modality-specific experts in vision-language MoEs. [13] introduced the expert ecology taxonomy (PURE_CORE, BROAD_CORE, WEAK_CORE, EDGE, NOISE, DEAD) and proposed the E formula. Our work extends this by providing cross-modal validation (language models), documenting expert revival, and establishing the task complexity effect.

3 Method

3.1 Architecture

We use a Hierarchical Mixture-of-Experts with three architectural tiers:

- **Encoder:** WideResNet-28-10 [12] for vision experiments, producing 256-dimensional features; GPT-2 style Transformer (8 layers, 512 hidden, 8 heads) with RoPE for language experiments.
- **16 Expert Worlds** organized into three tiers: 8 T0 (attribute-level), 4 T1 (logic-level), 4 T2 (top-level). For language, 4 of 8 Transformer layers contain 16-expert MoE FFN sub-layers.
- **Prototype vectors:** Each expert w has a learnable prototype $\mathbf{p}_w \in \mathbb{R}^{256}$ and a linear classifier head.
- **Router:** A linear layer $\mathbf{W}_r \in \mathbb{R}^{256 \times N}$ mapping encoded features to routing logits, with top-2 gating.

The hierarchical structure compresses router competition: experts within the same tier compete for overlapping input regions, but the tiered architecture reduces the effective competition set size, serving as a secondary stability factor alongside E.

3.2 The E Formula

The exploration budget E combines four hyperparameters:

$$E = \frac{T \cdot H}{O + B} \tag{2}$$

where T is the routing softmax temperature (higher $T \rightarrow$ softer routing, more exploration), H is the routing entropy loss weight (encouraging the router to maintain uncertainty), O is the oracle supervision weight (pushing the router toward a teacher signal), and B is the load balance loss weight (KL divergence from uniform routing).

E captures a fundamental tradeoff: *exploration vs. exploitation in routing space*. T and H promote exploration (trying different expert assignments). O and B promote exploitation (committing to known-good or uniform assignments). When E is too low, the router commits prematurely, locking in a winner-take-all dynamic from which some experts never recover.

We propose $E_{\text{crit}} \approx 0.5$ as the approximate critical threshold based on empirical observations across datasets. This threshold likely shifts with task complexity (Section 4.6).

3.3 Expert Ecology Taxonomy

Following [13], we classify each expert into one of six categories using its test-set usage (U) and accuracy (A):

Category	Usage	Accuracy
PURE_CORE	$\geq 3\%$	$\geq 50\%$
BROAD_CORE	$\geq 3\%$	30–50%
WEAK_CORE	$\geq 3\%$	$< 30\%$
EDGE	1–3%	$\geq 25\%$
NOISE	1–3%	$< 25\%$
DEAD	$< 1\%$	any

This taxonomy requires no hyperparameters—only usage and accuracy measured during a standard evaluation pass.

3.4 Training Protocol

All experiments share a common protocol:

- **Random warmup:** 15 epochs of random routing to initialize expert representations without bias.
- **Temperature annealing:** T follows a cosine schedule from T_{init} to $T_{\text{end}} = 0.3$ over 30 epochs.
- **Optimizer:** AdamW with decoupled weight decay (5×10^{-4}), learning rates 10^{-4} (encoder) and 10^{-3} (experts + router), cosine annealing to zero.
- **Batch size:** 128 for vision, 64 for language.
- **Evaluation:** Every 10 epochs, measuring Top-1/Top-2 accuracy (vision) or perplexity (language), plus per-expert usage and accuracy for ecology classification.

3.5 Experimental Design

We conduct 12 controlled experiments across two modalities:

Vision (8 experiments):

- CIFAR-100 phase map: 4 E values $\times$ 100 epochs at ortho=0
- CIFAR-100 ortho sweep: 5 ortho values $\times$ 100 epochs at $E = 0.545$
- CIFAR-10 low-E test: $E = 0.30$ vs. $E = 0.545$, 100 epochs
- CIFAR-10 ortho sweep: 5 ortho values $\times$ 100 epochs at $E = 0.545$
- CIFAR-10 E$\times$ortho interaction: $E = 0.30$ + ortho=0.10, 100 epochs
- TinyImageNet-200: $E = 0.545$, 100 epochs (16 experts)
- TinyImageNet-200: $E = 1.0$, 100 epochs (16 experts)
- TinyImageNet-200: $E = 0.545$, 100 epochs (32 experts)

Language (4 experiments):

- WikiText-2 BPE: GPT-2 MoE, $E = 0.545$, 100 epochs
- WikiText-2 ortho sweep: 5 ortho values $\times$ 100 epochs at $E = 0.545$
- WikiText-103 random-block: GPT-2 MoE, $E = 0.545$, 100 epochs
- No-warmup ablation: 4 conditions $\times$ 100 epochs on CIFAR-100

All experiments use no external intervention (no Claude API, no manual tuning) during training. The Claude-in-the-loop methodology [13] was used only for initial diagnosis and hypothesis generation, not for the experiments reported here.

4 Results

4.1 E Controls Ecosystem Quality, Not Survival (Phase Map)

Table 1 shows the four-value phase boundary mapping on CIFAR-100 at $\lambda_o = 0$. All four E values spanning a $4.6\times$ range eventually achieve zero dead experts. E governs *how many* experts become high-quality, not whether the ecosystem survives.

This is the E–PURE gradient: higher E produces more PURE_CORE experts independent of survival. Accuracy shows diminishing returns beyond $E = 0.35$, suggesting that ensemble accuracy and ecological richness are partially decoupled.

Table 1: Phase boundary mapping at $\lambda_o = 0$, 100 epochs, CIFAR-100

E	T	H	Top-2	PURE_CORE
0.14	1.0	0.05	67.87%	7
0.35	2.0	0.07	67.60%	10
0.44	2.5	0.08	66.72%	11
0.65	3.0	0.12	68.02%	**14**

4.2 E $\geq$ 0.5 Guarantees DEAD=0 (Ortho Sweeps)

Table 2 consolidates ortho sweeps across three datasets at $E = 0.545$. The central finding: at sufficient E, ortho from 0 to 0.20 produces DEAD=0 across all tested datasets.

Table 2: Ortho sweeps at $E = 0.545$: DEAD=0 for all ortho values

Dataset	DEAD count at ortho=				
	0.00	0.02	0.05	0.10	0.20
CIFAR-100	0	0	0	0	0
CIFAR-10	0	0	0	0	0
WikiText-2	0	0	0	0	0

The ortho toxicity effect (increased dead expert count) observed at low $E = 0.09$ [15] does not generalize to healthy E. Ortho toxicity is E-dependent and dataset-dependent, not universal.

4.3 Cross-Modal: E Framework Generalizes to Language

Table 3 shows that $E = 0.545$ produces DEAD=0 across language modeling experiments using a GPT-2 style Transformer with MoE FFN layers (16 experts, top-2 gating).

Table 3: Language model results at $E = 0.545$

Dataset	ortho	PPL	DEAD	Active
WikiText-2 BPE	0.00	35041	0	16/16
WikiText-2 BPE	0.02	33493	0	16/16
WikiText-2 BPE	0.05	35737	0	16/16
WikiText-2 BPE	0.10	33163	0	16/16
WikiText-2 BPE	0.20	37812	0	16/16
WikiText-103	0.00	6918	0	16/16

The E framework transfers cleanly from vision to language: same E threshold, same ecology patterns, same ortho tolerance. This cross-modal consistency supports the generality of E as a control parameter.

4.4 Low-E Collapse and Phase Distinction

Table 4 shows the effect of lowering E below the critical threshold on CIFAR-10.

Crucially, the collapse at moderately low E is *reversible*—experts recover when E is restored. We distinguish three ecological phases:

1. **Healthy** ($E \geq 0.5$): DEAD=0 sustained throughout training.
2. **Reversible sub-health** ($0.2 < E < 0.5$): DEAD > 0 but experts recover when E is restored. The balance-driven recovery mechanism (Section 4.7) remains functional.

Table 4: CIFAR-10 low-E experiment: phase distinction

E	T	H	O	B	Top-1	DEAD
0.545	3.0	0.10	0.15	0.40	87.47%	**0**
0.300	3.0	0.10	0.15	0.85	84.58%	**2**

3. **Irreversible collapse** ($E < 0.2$, predicted): Large-scale permanent death. The rich-get-richer feedback loop overwhelms the revival mechanism.

This phase distinction mirrors critical phenomena: near the critical point, the system is sub-healthy but self-repairing; far below it, the feedback loop becomes irreversible.

4.5 E×Ortho Interaction at Low E

Adding ortho=0.10 at $E = 0.30$ does *not* amplify toxicity (DEAD=1 vs. DEAD=2 for ortho=0 baseline). This confirms that ortho is not a primary driver of expert death—E is. Ortho amplifies collapse only when E is already insufficient, consistent with the interaction model proposed in [15].

4.6 Task Complexity Shifts the Critical E Threshold

Table 5 reveals that task complexity (number of fine-grained classes) shifts the effective E threshold.

Table 5: Task complexity effect at $E = 0.545$

Dataset	Classes	Top-1	DEAD
CIFAR-10	10	89.59%	**0**
CIFAR-100	100	68.02%	**0**
TinyImageNet	200	38.15%	**7** (E50)

CIFAR-10 (10 classes) and CIFAR-100 (100 classes) maintain DEAD=0 at $E = 0.545$. TinyImageNet-200 (200 visually similar classes: dog breeds, vehicles, insects) produces DEAD=7 at E50, recovering to DEAD=4 by E80. This is not a failure of E but a revelation: **task complexity is a missing variable in the E formula**. The critical E threshold rises with the number of fine-grained categories.

Two controlled experiments test whether the TinyImageNet collapse can be fixed by adjusting E or expert capacity:

Table 6: Attempted fixes for TinyImageNet-200 collapse

Configuration	E	Experts	Top-1 (E40)	DEAD (E40)
Baseline	0.545	16	33.41%	7
Raise E	1.000	16	27.19%	7
More experts	0.545	32	26.99%	**25**

Raising E to 1.0 fails. Doubling the exploration budget produces nearly identical ecology (DEAD=7 at E40) and slightly worse accuracy (27.19% vs. 33.41%), confirming that the TinyImageNet collapse is *not* caused by insufficient router exploration. The task complexity effect is orthogonal to E.

Increasing experts to 32 backfires. Doubling expert count from 16 to 32 worsens collapse dramatically (DEAD=25/32 vs. DEAD=7/16 at E40). The expanded capacity creates *more* competition, not less—without a mechanism to structure how 200 classes map to 32 experts, the additional capacity only deepens the rich-get-richer spiral.

These negative results sharpen the central claim: for complex datasets, neither raising E nor increasing expert count is sufficient. The required intervention is likely structural—either a hierarchical decomposition that reduces the effective competition set per expert, or a curriculum that gradually introduces fine-grained distinctions.

This implies that for complex datasets, increasing E alone is insufficient—either more experts or a different architectural prior is needed.

4.7 Expert Revival: Dead Experts Can Resuscitate

Traditional MoE theory (Switch Transformer, GShard) holds that dead experts are *permanently* dead—a rich-get-richer feedback loop: no tokens → no gradient → worse representations → even fewer tokens. Load balancing can *prevent* death but cannot *reverse* it.

We observe the opposite on TinyImageNet-200 at $E = 0.545$:

Table 7: Expert revival trajectory: TinyImageNet-200, $E = 0.545$

Epoch	DEAD	Active
E0	4	14/16
E10	**12**	12/16
E20	8	10/16
E30	7	15/16
E40	7	15/16
E50	7	16/16
E60	5	16/16
E70	5	16/16
E80	**4**	16/16

DEAD count peaks at 12 at E10, then gradually falls to 4 by E80—**8 experts resuscitated**. This reversal is possible because of the **balance loss** ($B = 0.40$), which combines a KL-divergence term on routing importance distribution with a variance penalty on expert assignment counts. Together, these terms create sustained pressure toward distributed routing: the router is penalized for concentrating flow on a subset of experts, forcing periodic re-exploration of inactive experts. Dead experts benefit when the balance loss forces the router to re-explore them; they receive task gradient, their predictions improve, and they attract organic routing flow. A systematic six-condition ablation study (reported separately in [14]) identifies balance loss as the sole essential mechanism for expert revival. Four other tested mechanisms—divergence loss, temperature annealing, routing entropy, and prototype vectors—are individually non-essential; revival proceeds in all conditions where balance loss is present.

This is a meaningful departure from traditional MoE theory: dead experts are not permanently lost capacity. With adequate load-balancing pressure, the death spiral is reversible.

4.8 Overfitting Is Decoupled from Expert Ecology

WikiText-103 training shows severe perplexity divergence (overfitting on the small relative dataset) while expert ecology remains fully intact: DEAD=0, 16/16 experts active throughout training. This demonstrates that **model overfitting and expert ecological health are independent axes.**

Expert death is driven by routing dynamics (E and ortho), not by whether the model is overfitting or underfitting. A model can overfit severely while maintaining a perfectly healthy expert ecology. Conversely (as shown by the low-E experiments), a model can generalize well while experiencing ecological collapse. These are orthogonal diagnostic dimensions.

4.9 Warmup Is Optional

A four-condition ablation on CIFAR-100 tests whether random routing warmup is required for ecological recovery:

Table 8: No-warmup ablation: all conditions achieve DEAD=0

Condition	Top-2	PURE	DEAD=0 by
Warmup + ortho=0	67.55%	11	E80
Warmup + ortho=0.1	66.65%	12	E70
No-warmup + ortho=0	66.86%	11	**E60**
No-warmup + ortho=0.1	65.82%	11	E70

All four conditions achieve DEAD=0. No-warmup conditions actually show *milder* collapse (peak DEAD=9 vs. 13 with warmup) and *faster* recovery (E60 vs. E80). Random routing at the start may confuse the router when it transitions to real routing. The only necessary conditions for a healthy ecology are $E \geq 0.5$ and $\lambda_o \leq 0.1$. Warmup is optional.

4.10 Tier-Level Ecological Dynamics: Collapse, Self-Loop, and Hierarchy

To understand *which* experts collapse and *how* routing flows between tiers, we instrument a per-tier analysis at the evaluation checkpoint. For each test sample, we record the top-1 expert choice and classify each expert into its architectural tier (T0: foundation, T1: logic, T2: fine-grained). We define easy samples as those with ensemble confidence $p > 0.7$ and hard samples as $p < 0.4$.

Table 9 shows per-tier ecology at the most mature evaluation points for both TinyImageNet-200 experiments.

Table 9: Per-tier ecological state at maturity (32e E50, 16e E30)

	32e $E = 0.545$ (E50)			16e $E = 1.0$ (E30)		
	T0(16)	T1(8)	T2(8)	T0(8)	T1(4)	T2(4)
Usage %	43.5	29.3	27.2	82.9	2.2	14.9
Active	2/16	3/8	2/8	4/8	1/4	3/4
Hard %	7.2	–	5.4	12.0	–	8.2
Easy %	66.1	–	69.0	53.6	–	60.1

The middle tier is permanently starved. In both configurations, T1 receives negligible routing flow—2.2% in the 16e case, with only 1 of 4 experts above the activity threshold. This is not a transient effect; the T1 collapse is stable across all evaluation temperatures and persists from early to late training. Three-tier hierarchical MoE spontaneously collapses into a *two-tier functional structure*, with the middle tier squeezed out by competition from above and below. This suggests that for 200-class tasks, the effective hierarchy depth is 2, not 3—the additional tier is architectural redundancy that routing dynamics naturally eliminate.

T0 self-loop blocks upward flow. Table 10 reports the routing transition matrix: for each sample, we record the tier of the top-1 expert (T_{first}) and the tier of the top-2 expert (T_{second}).

Table 10: Routing flow matrix: $T_{\text{first}} \rightarrow T_{\text{second}}$ (% of total samples)

	32e E50			16e E30		
$T_{\text{first}} \rightarrow$	T0	T1	T2	T0	T1	T2
T0→	17.5	12.7	13.2	**65.3**	6.0	11.6
T1→	15.2	3.7	10.5	1.6	0.0	0.6
T2→	14.3	7.9	5.1	9.7	1.5	3.7

In the 16e case, 65.3% of samples whose top-1 choice is a T0 expert also have their top-2 choice in T0—the flow *circulates within the bottom tier*. Only 5.7% of T0-routed samples reach T2 as the second choice. This T0 self-loop indicates that the router cannot identify complementary high-tier experts for the samples assigned to T0, which is precisely the "difficulty gating failure" predicted when task complexity exceeds the architecture's tier differentiation capacity.

In the 32e case, the flow pattern is more distributed but the absolute number of active experts is even lower (7/32 vs. 8/16). Doubling expert count while keeping the tier ratio fixed (16:8:8) dilutes per-expert usage without creating effective specialization pressure.

Hard sample gradient: T0 consistently absorbs harder samples than T2. Across all evaluation temperatures (0.1–5.0), the T0 hard-sample ratio exceeds T2 by a factor of 1.2×–1.6×. This gradient is *structurally stable*: it appears in both the 32e and 16e configurations, at all temperature settings, and in both the healthy (DEAD=0) and collapsed regimes observed. Lower-tier experts inherently undertake harder, more ambiguous samples, forming a hierarchical difficulty division mechanism. The existence of this gradient even in heavily collapsed models (where accuracy is near zero) suggests it is a property of the routing geometry, not of task performance.

4.11 Ecological Structure Is Temperature-Invariant

A methodological concern is whether off-line ecological analysis (at a fixed evaluation temperature) faithfully reflects in-training ecological states, since training uses a cosine-annealed temperature schedule. To test this, we run a temperature sensitivity scan on both models, evaluating at seven temperatures spanning $T \in [0.1, 5.0]$ while holding all model weights fixed.

Table 11: Temperature sensitivity scan: 32e E50 and 16e E30

	32e E50			16e E30		
T	Acc%	T0/T2 hard	T0 act	Acc%	T0/T2 hard	T0 act
0.1	0.11	1.2×	2	0.12	1.3×	4
0.3	0.11	1.3×	2	0.11	1.5×	4
0.5	0.11	1.3×	2	0.12	1.5×	4
0.7	0.11	1.3×	2	0.13	1.5×	4
1.0	0.10	1.4×	2	0.13	1.5×	4
2.0	0.12	1.3×	2	0.13	1.5×	4
5.0	0.12	1.3×	2	0.12	1.6×	4

Two findings emerge. First, **expert allocation is temperature-invariant**: tier usage percentages and active expert counts are *identical* across the full 50× temperature range. The router's softmax logits are so sharply peaked that temperature scaling does not alter the top-2 selection. This means the routing has entered a *lockdown state*—a deterministic, near-one-hot assignment pattern that persists regardless of the exploration budget.

Second, **the hard-sample gradient across tiers is temperature-invariant**. While the absolute hard-ratio percentage drifts upward with temperature (as higher temperature spreads confidence mass, pushing more samples below the 0.4 threshold), the T0/T2 *ratio* remains stable at 1.2×–1.6×. The hierarchical difficulty division is a structural property of the routing geometry, not an artifact of the evaluation temperature.

Implication for ecological assessment. The temperature invariance of tier-level ecological metrics (usage, active count, hard-ratio gradient) means that off-line ecological diagnostics at a single temperature are valid. However, *absolute accuracy* measured at a fixed temperature is **not** representative of in-training accuracy, because the training evaluation uses the current annealed temperature. Ecological structure and task performance must be assessed separately: the former is temperature-invariant; the latter is not.

5 Discussion

E as a control parameter. The quantity $E = T \cdot H/(O + B)$ serves a role analogous to the Reynolds number in fluid dynamics: a single dimensionless quantity that predicts regime transitions in a complex dynamical system. Just as Re predicts the laminar-to-turbulent transition, E predicts the healthy-to-collapsed transition in expert ecology. The critical value $E_{\text{crit}} \approx 0.5$ is approximate and task-dependent, but the *concept* of a unified exploration budget is general.

Practical engineering implications. The finding that $E \geq 0.5$ alone guarantees DEAD=0 removes the need for traditional load-balancing auxiliary losses. This is a significant simplification: rather than tuning multiple balancing hyperparameters (load balance weight, capacity factor, expert dropout rate, minimum usage), practitioners need only ensure E is sufficient. The recipe: (1) set $\lambda_o = 0$, (2) choose T and H such that $E \geq 0.5$, (3) let experts self-organize. If expert specialization is insufficient, mild ortho (0.02–0.05) can be introduced *after* verifying ecological stability.

Task complexity as a missing variable. The TinyImageNet-200 results reveal that the critical E threshold rises with the number of fine-grained classes. The current E formula treats all datasets as equivalent. A natural extension is:

$$E_{\text{eff}} = \frac{T \cdot H}{(O + B) \cdot f(C)} \tag{3}$$

where $f(C)$ is a task complexity function that increases with the number of classes C. Candidates include $f(C) = \log C$, $f(C) = \sqrt{C}$, or a learned function. This refinement is left to future work.

Expert revival mechanism. The observation that dead experts can resuscitate is, to our knowledge, the first documented case of reversal of the MoE death spiral. A systematic ablation study (reported separately in [14]) identifies balance loss (weight 0.40) as the sole essential mechanism: without balance loss, DEAD stalls permanently at 12 with zero recovery over 80 consecutive epochs. With balance loss present, revival proceeds regardless of which other component (divergence loss, temperature annealing, routing entropy, prototype vectors) is removed. The key principle is **routing diversity enforcement**: any mechanism that prevents the router from permanent concentration on a subset of experts (balance loss, expert-choice routing, scheduled expert dropout) can potentially enable revival.

Hierarchical structure as stability factor. The three-tier hierarchy reduces the effective size of the competition set at each level, acting as a secondary stability factor. Rather than all N experts competing globally, competition is structured: T0 experts compete primarily with other T0 experts, T1 with T1, etc. This reduces the winner-take-all pressure that drives collapse. Hierarchy and E are complementary: E provides the exploration budget; hierarchy reduces the need for it.

Overfitting–ecology decoupling. The decoupling of overfitting from ecology is practically significant: it means that expert ecology can be diagnosed and treated independently of model generalization. A model suffering from dead experts is not necessarily overfitting; a model that is overfitting does not necessarily have ecological problems. These are orthogonal axes in model health.

Tier collapse: three tiers degenerate to two. Across both TinyImageNet-200 experiments, the middle tier (T1) is permanently starved—receiving 2.2% usage in the 16e case and showing 5 of 8 experts dead in the 32e case. This is not a malfunction but a revealing structural phenomenon: **three-tier hierarchical MoE spontaneously compresses into a two-tier functional structure under high task complexity**. The router effectively decides that the middle tier is redundant, routing samples to either the foundation tier (T0) or the fine-grained tier (T2). This suggests that the effective hierarchy depth is determined not by the architectural design but by the interaction between task complexity and routing dynamics. For 200-class fine-grained classification, the optimal functional depth appears to be 2, not 3. This is, to our knowledge, the first empirical documentation of spontaneous tier collapse in hierarchical MoE.

Routing lockdown and T0 self-loop. The temperature scan (Section 4.11) reveals that the router enters a *lockdown state*: expert allocation is completely invariant across a 50× temperature range (0.1–5.0). The softmax logits are so sharply peaked that temperature scaling cannot alter the top-2 selection. Combined with the 65.3% T0→T0 self-loop rate (Table 10), this paints a picture of **complexity-induced routing obstruction**: when the task is too fine-grained for the architecture's tier differentiation capacity, the router cannot identify meaningful cross-tier expert complementarity. Samples assigned to T0 circulate within T0; the upward routing pathway is blocked. This routing lockdown is the dynamical mechanism underlying the task complexity effect identified in Section 4.6—it explains *why* raising E fails: the exploration budget cannot unlock routing pathways that the architecture cannot structurally differentiate.

Ecological structure as a performance-decoupled invariant. The temperature scan establishes that tier-level ecological metrics—usage distribution, active expert count, and the hard-sample gradient across tiers—are *invariant* to evaluation temperature. However, absolute accuracy is not: off-line evaluation accuracy ($\sim$0.1%) differs from in-training evaluation accuracy (23–35%) by two orders of magnitude in the collapsed models. This decoupling has a methodological implication: **ecological diagnostics** (DEAD count, tier usage, hard-ratio gradient) can be reliably assessed at any reasonable temperature, but **performance diagnostics** (accuracy, perplexity) must use the same temperature as the training evaluation. The two classes of metrics are orthogonal and must not be conflated in analysis.

Limitations. We test two architecture families (WideResNet encoder + MoE heads; GPT-2 Transformer + MoE FFN) with 16–32 experts. The specific E_{crit} threshold likely depends on model scale, number of experts, and data complexity. We do not test very large models (100+ experts, billion-parameter scale), where additional stability mechanisms may emerge. All training uses a random routing warmup of 15 epochs; while we show warmup is optional, its interaction with very low E regimes is not fully explored. The task complexity function $f(C)$ remains unspecified.

6 Conclusion

We have introduced $E = T \cdot H/(O + B)$, a dimensionless control parameter for Mixture-of-Experts ecology, and validated it across 12 controlled experiments spanning two modalities and four datasets. The core empirical finding is that $E \geq 0.5$ alone is sufficient to guarantee zero dead experts, removing the need for handcrafted load-balancing auxiliary losses.

Five additional findings enrich the theoretical picture: (1) dead experts can resuscitate, contrary to traditional MoE theory; (2) ortho toxicity is dataset-dependent, not universal; (3) task complexity shifts the critical E threshold; (4) model overfitting is decoupled from expert ecological health; (5) the hierarchical tier structure is a secondary stability factor.

The broader implication is that expert ecology in MoE models is not a pathology to be suppressed but a dynamical system to be understood. The E parameter provides a compact diagnostic for navigating that system.

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